COMSATS University Islamabad, Vehari Campus

Project Proposal

(SCOPE DOCUMENT)

for

PROJECT NAME

Ghost Interview (AI Mock Interviewer)

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SCOPE DOCUMENT REVSION HISTORY

No.	Comment	Action

Supervisor Signature:_____

Date:_____

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Project Category: (Select all the major domains of proposed project**)**

○ **A-**Desktop Application/Information System ■**B-**Web Application/Web Application based Information System ○**C-** Problem Solving and Artificial Intelligence ○ **D-**Simulation and Modeling ○ **E-** Smartphone Application ○ **F-**Smartphone Game ○ **G-** Networks ○ **H-** Image Processing ○Other (specify category) _____

Abstract

AI Interview Ghost (Mock Interviewer) is an intelligent web-based interview preparation system designed to simulate real-time job interviews using artificial intelligence. The system provides users with an interactive interview experience through a virtual AI avatar that asks domain-specific interview questions using a realistic human-like voice. Unlike traditional mock interview systems, this project focuses not only on the verbal responses of the user but also on non-verbal cues such as facial expressions, eye contact, confidence level, and emotional state.

The proposed system analyzes the candidate's speech, emotions, and body language using advanced AI models and generates a comprehensive Confidence and Competency Report. This report highlights communication skills, nervousness indicators, usage of filler words, and overall interview readiness. The system aims to help students and job seekers improve their interview performance through data-driven feedback and self-assessment.

1. Introduction

This document presents the project scope for the proposed system titled *AI Interview Ghost (Mock Interviewer)*. The purpose of this project is to provide an intelligent and interactive interview preparation platform using artificial intelligence technologies. Interviews are a critical part of job selection processes, and many candidates struggle due to lack of practice and real-time feedback. Traditional mock interviews are limited in availability and often lack detailed performance analysis.

With the advancement of AI, it is now possible to simulate realistic interview scenarios. This system integrates voice interaction, facial emotion detection, and speech analysis to create a virtual interviewer experience. The AI avatar interacts with users through voice-based questions and evaluates their responses using multiple AI models. The project is developed as a web-based application to ensure accessibility and scalability. The system is particularly useful for students, fresh graduates, and professionals preparing for technical or HR interviews.

2. Problem Statement

Many job candidates face difficulty during interviews due to nervousness, lack of confidence, and poor communication skills. Most available interview preparation platforms focus only on question-answer practice and ignore non-verbal communication aspects such as facial expressions, eye contact, and emotional stability. Human mock interviews are costly, time-limited, and not always available to everyone.

Existing systems also fail to provide real-time emotion analysis and speech behavior insights such as filler word usage. Candidates do not receive objective feedback on their confidence and emotional performance. There is a lack of intelligent systems that combine voice interaction, emotion detection, and AI-driven evaluation in a single platform. This project is developed to overcome these limitations and to provide a comprehensive interview preparation solution while enhancing the students' skills in artificial intelligence, machine learning, and web technologies.

3. Problem Solution for Proposed System

The proposed AI Interview Ghost system addresses the identified problems by providing a fully automated mock interview environment. The system allows users to select a job role, after which an AI avatar conducts an interview using voice-based interaction. The user responds through a webcam and microphone, enabling the system to analyze both verbal and non-verbal behavior.

Speech-to-text conversion is used to analyze spoken responses and detect filler words. Emotion detection models analyze facial expressions and eye movement to determine nervousness and confidence levels. Sentiment analysis is applied to evaluate the emotional tone of the answers. Based on these analyses, the system generates a detailed Confidence and Competency Report. This solution provides users with actionable insights and helps them improve interview performance through repeated practice.

4. Related System Analysis/Literature Review

System 1: HireVue
HireVue provides AI-based video interviews but lacks detailed emotional feedback and real-time guidance.

System 2: Pramp
Pramp offers peer-to-peer mock interviews but depends on human availability and lacks AI-based emotion analysis.

System 3: Interviewing.io
Interviewing.io focuses on technical interviews but does not analyze facial expressions or emotional behavior.

Table 1: Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
HireVue	Limited emotion insights	Deep emotion & confidence analysis
Pramp	Human dependency	Fully automated AI interviewer
Interviewing.io	No emotion detection	Facial and sentiment analysis

5. Advantages/Benefits of Proposed System

- Real-time AI-based mock interview experience
- Emotion and confidence analysis using webcam
- Detection of filler words and speech patterns
- Detailed performance report generation
- Accessible web-based system
- Cost-effective alternative to human interviews
- Improves communication and interview confidence

6. Scope

The scope of AI Interview Ghost includes developing a web-based interview simulation system using AI technologies. The system will allow users to select job roles and participate in AI-driven interviews. The AI avatar will ask interview questions using voice interaction. The system will analyze user responses through speech-to-text processing and facial emotion detection. A confidence and competency report will be generated after each interview session.

The project includes sentiment analysis, filler word detection, and emotion recognition. The system will not provide real job placements or human interview scheduling. Advanced AI model training is out of scope and pre-trained models will be used. The focus remains on interview preparation and performance analysis.

7. Modules

Module 1: User Role Selection Module

Allows users to select job roles and interview categories.

Module 2: AI Interview Avatar Module

Conducts interviews using voice-based AI avatar.

Module 3: Speech Analysis Module

Converts speech to text and detects filler words.

Module 4: Emotion & Sentiment Analysis Module

Analyzes facial expressions and emotional tone.

Module 5: Confidence & Competency Report Module

Generates performance reports with insights.

Module 6: System Dashboard Module

Displays interview history and analytics.

8. System Limitations/Constraints

- Requires stable internet connection
- Emotion detection accuracy depends on camera quality
- Limited to predefined job roles
- Privacy concerns related to webcam usage

9. Software Process Methodology

The Object-Oriented Software Development Methodology will be used for this project. This methodology is selected due to its modular structure and reusability. It supports scalability and effective system maintenance. It is suitable for AI-based web applications.

10. Tools and Technologies

Table 2: Tools and Technologies for Proposed Project

Tools	**Version**	**Rationale**
Visual Studio Code	Latest	Development IDE
React.js	18+	Frontend framework
Node.js	20+	Backend runtime
WebRTC	Latest	Camera & mic access

Technologies:

Technology	**Version**	**Rationale**
JavaScript	ES6+	Programming
OpenAI Whisper	Latest	Speech-to-text
Gemini API	Latest	Interview logic
DeepFace / Hume AI	Latest	Emotion detection
ElevenLabs	Latest	AI voice generation

11. Project Stakeholders and Roles

Table 3: Project Stakeholders for Proposed Project

Project Sponsor	COMSATS University, Islamabad
Stakeholder	● Students: System development ● Supervisor: Project guidance ● FYP Committee: Evaluation

12. Team Members Individual Tasks/Work Division

Table 4: Team Member Work Division for Proposed Project

Student Name	**Registration No**	**Responsibility**
Ehsan Hamza	SP23-BCS-019	AI models & backend
Shaharyar Altaf	SP23-BCS-055	Frontend & UI

13. Data Gathering Approach

Data will be gathered through interviews, online research, and analysis of existing interview platforms. Questionnaires will be used to understand user requirements. Research papers and AI documentation will also be reviewed.

14. Concepts

Concept 1: Artificial Intelligence
Used for interview logic and evaluation.

Concept 2: Emotion Detection
Facial expression analysis using AI models.

Concept 3: Speech-to-Text Processing
Converts user speech into text for analysis.

Concept 4: Sentiment Analysis
Evaluates emotional tone of responses

15. Gantt chart

The Gantt chart will be created using MS Project to define timelines, dependencies, and milestones for each module.

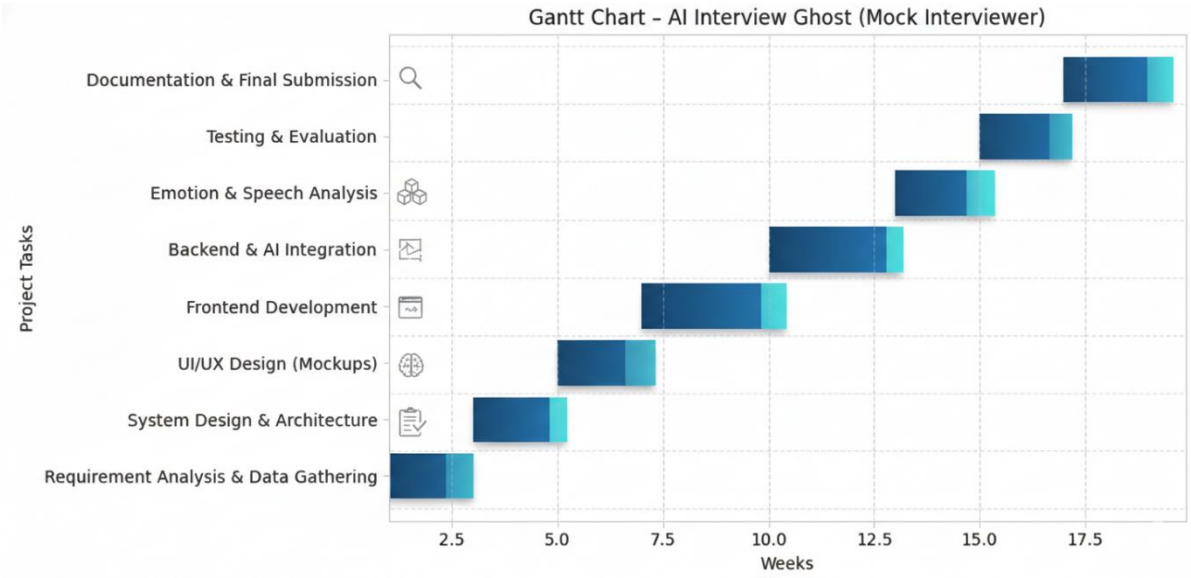


Figure 1: Gantt Chart

16. Mockups

Mockups will include interview screen, AI avatar interaction, report dashboard, and analytics view.



Figure 2: Job Role Selection Screen

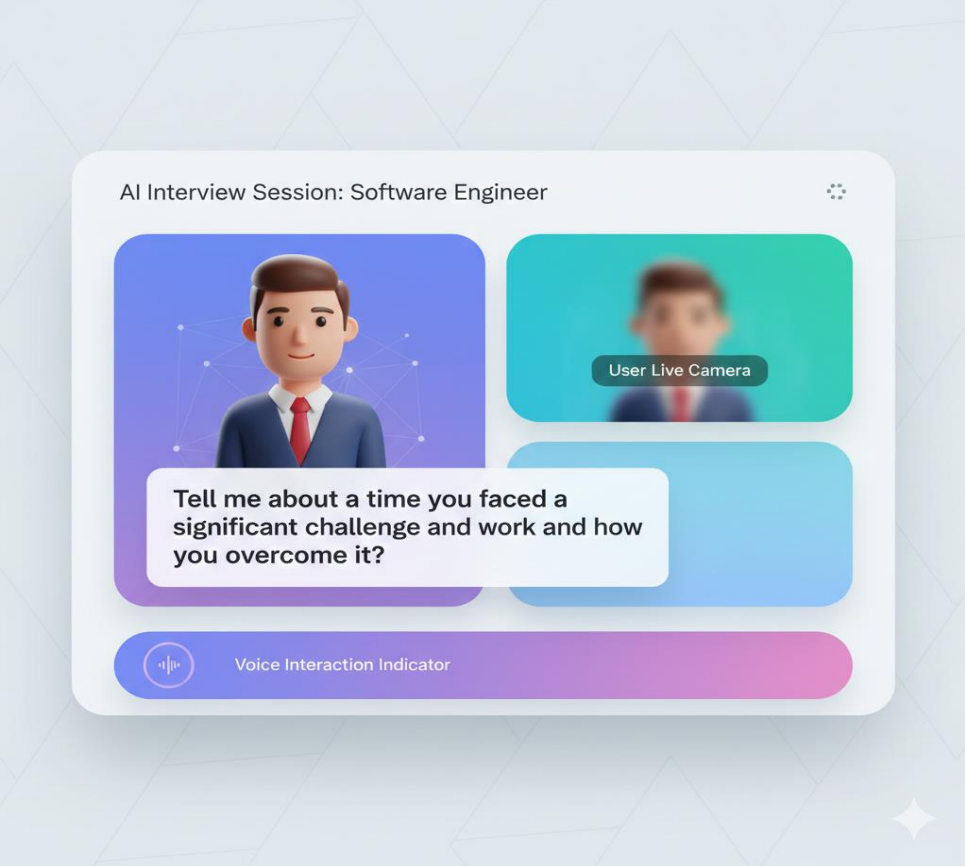


Figure 3: AI Interview Avatar Interaction Screen



Figure 4: Emotion and Speech Analysis Dashboard

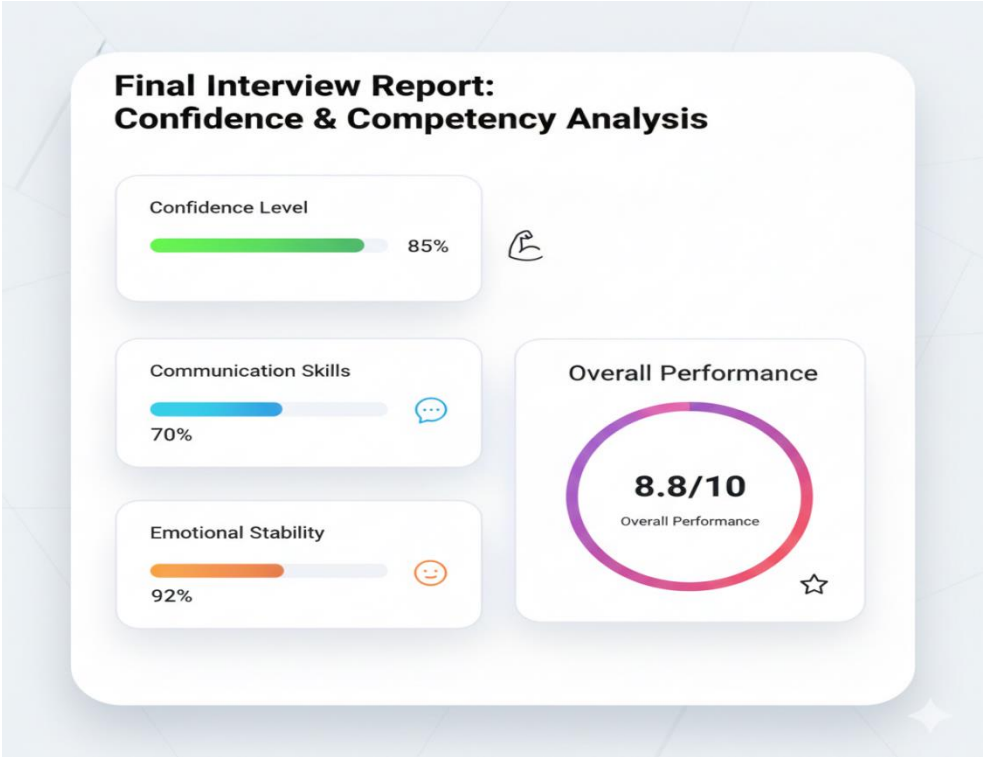


Figure 5: Confidence and Competency Report Screen

17. Conclusion

AI Interview Ghost is an innovative interview preparation system that combines AI-driven voice interaction, emotion analysis, and performance evaluation. The project provides a realistic interview environment and actionable feedback. It enhances user confidence and communication skills. The system demonstrates the practical application of artificial intelligence in real-world scenarios

References

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